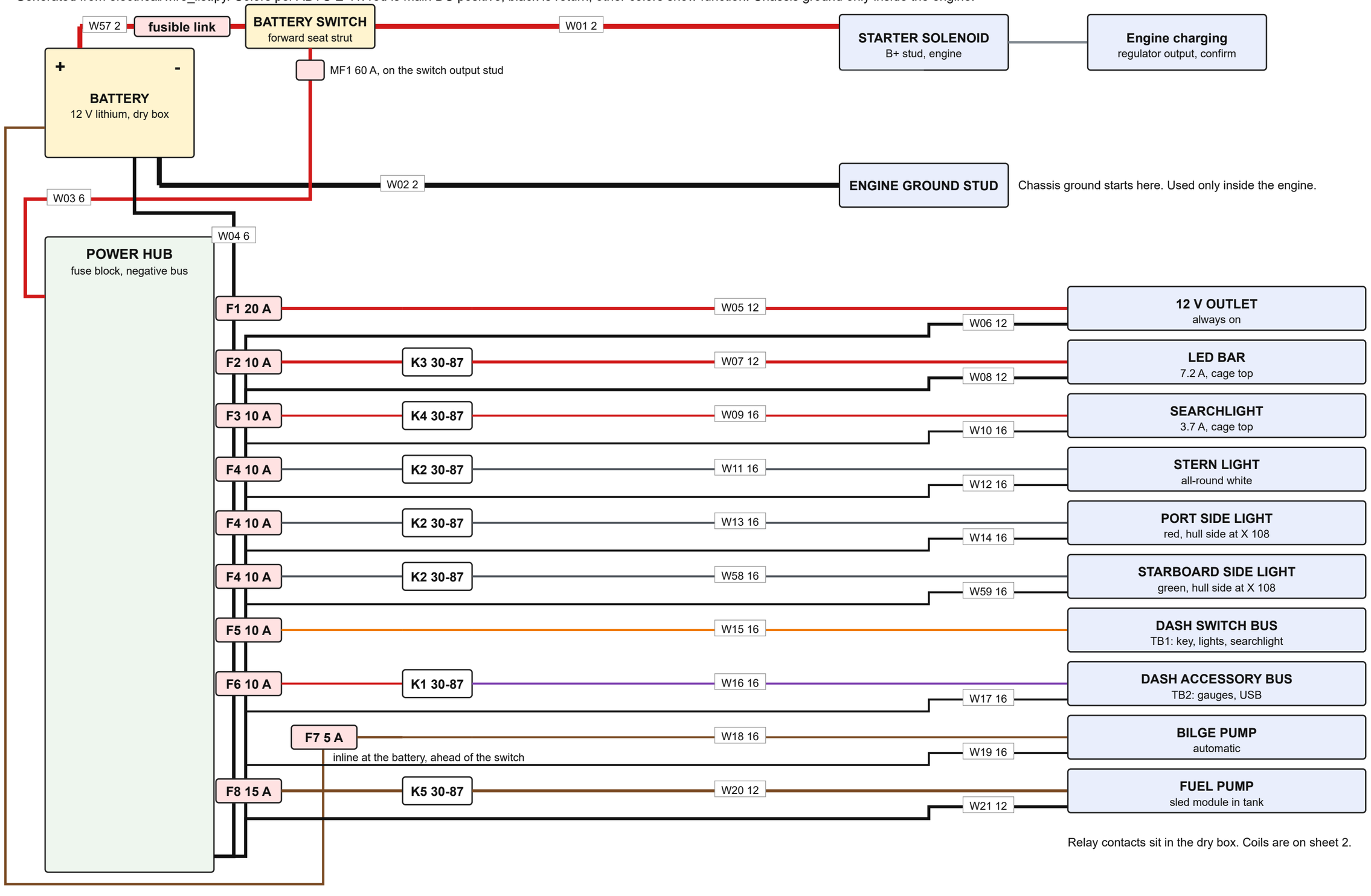
Minimal 1448 airboat electrical schematic: power distribution

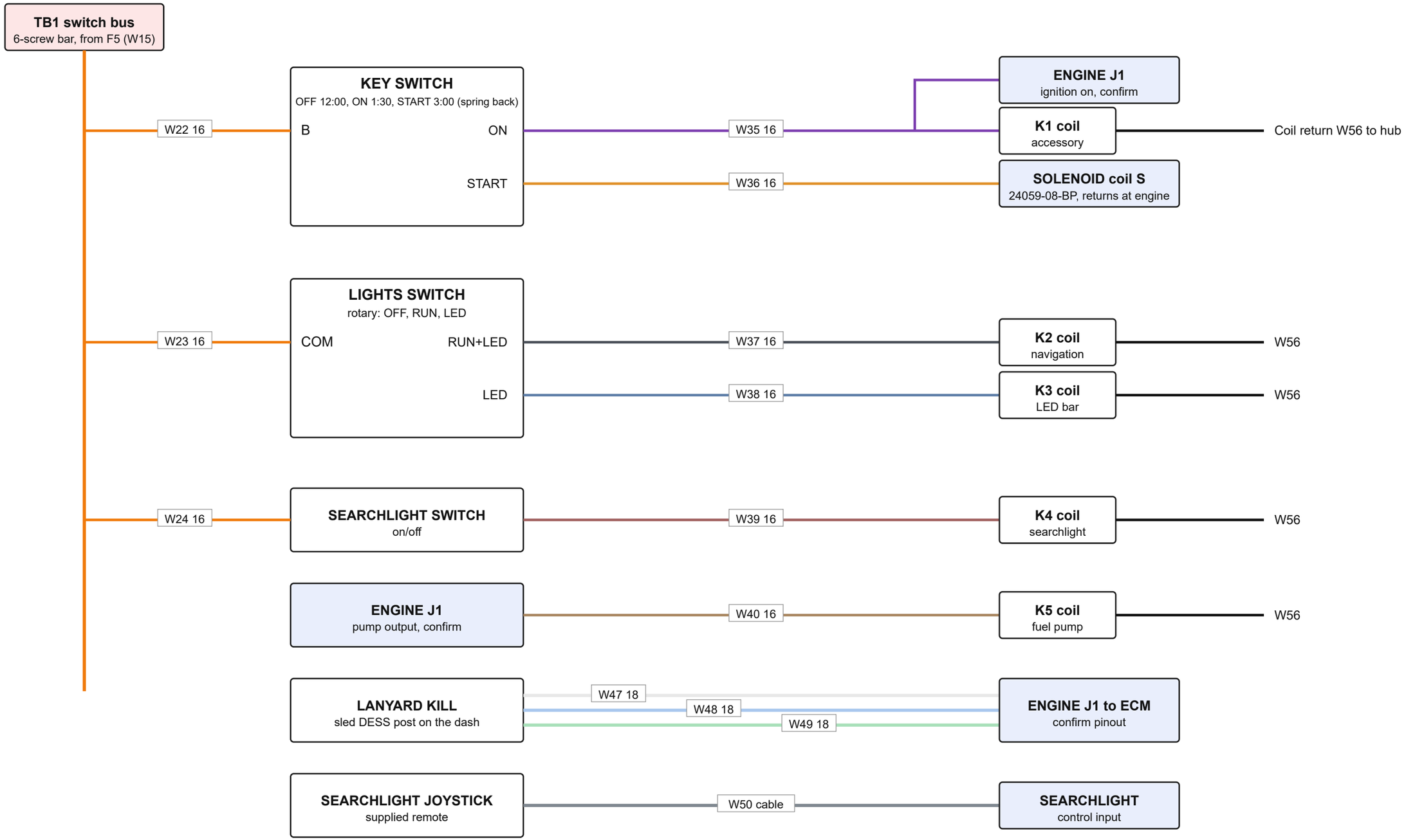
Generated from electrical/wire_list.py. Colors per ABYC E-11: red is main DC positive, black is return, other colors show function. Chassis ground only inside the engine.



Relay contacts sit in the dry box. Coils are on sheet 2.

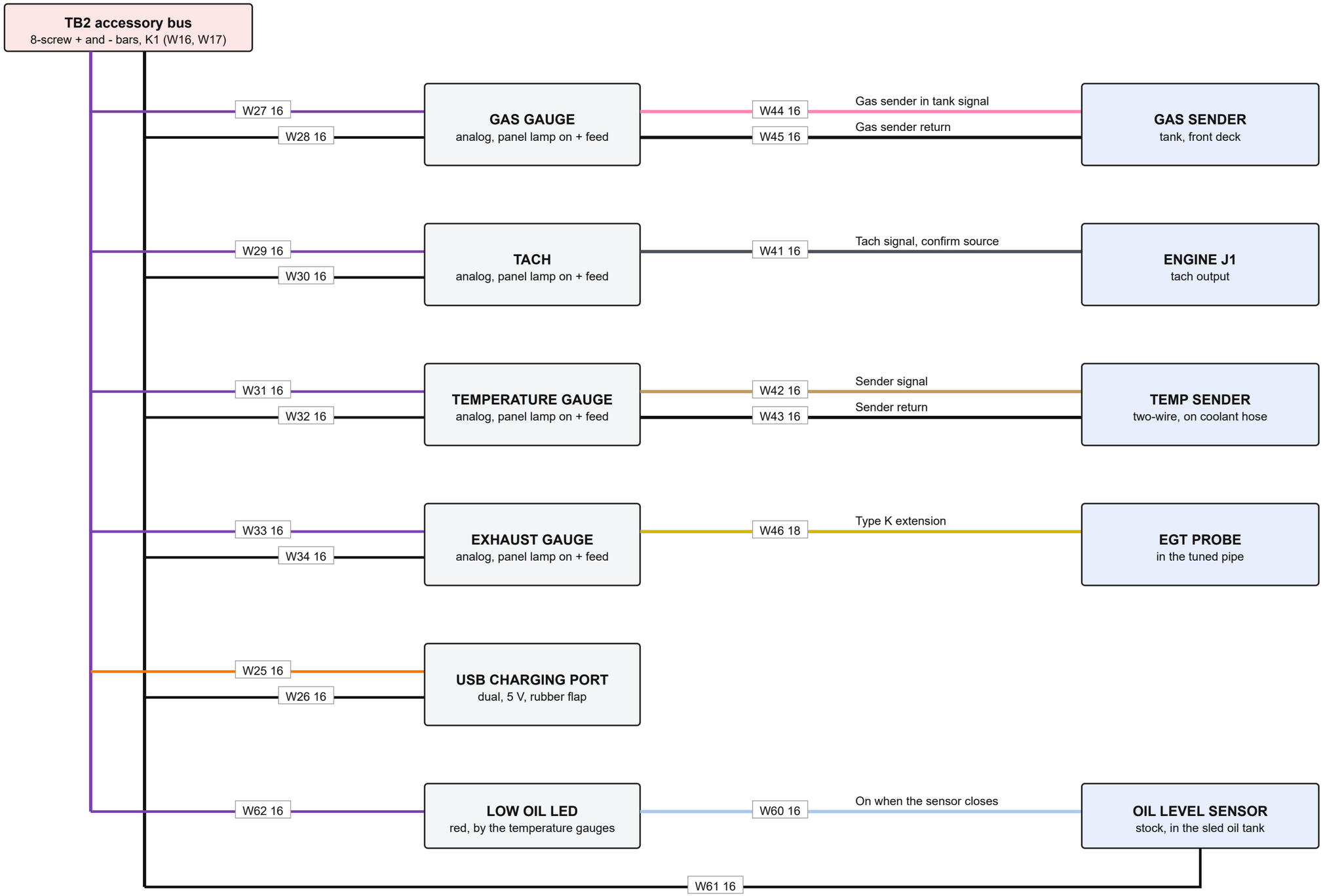
Minimal 1448 airboat electrical schematic: switching and signals

Generated from electrical/wire_list.py. Colors per ABYC E-11: red is main DC positive, black is return, other colors show function. Chassis ground only inside the engine.



Minimal 1448 airboat electrical schematic: gauges and sensors

Generated from electrical/wire_list.py. Colors per ABYC E-11: red is main DC positive, black is return, other colors show function. Chassis ground only inside the engine.



Minimal 1448 airboat electrical schematic: wire and fuse lists

Generated from electrical/wire_list.py. Colors per ABYC E-11: red is main DC positive, black is return, other colors show function. Chassis ground only inside the engine.

Wire	Function	AWG	Color	Amps	Ft	Drop
W01	Battery switch to starter solenoid	2	red	150	7.1	
W02	Battery negative to engine ground	2	black	150	7.4	
W03	Battery switch to hub	6	red	45	1.3	
W04	Battery negative to hub	6	black	45	2.1	
W05	F1 to 12 V outlet	12	red	20	5.7	2.7%
W06	12 V outlet return	12	black	20	5.7	
W07	K3 to LED bar	12	red	7.2	16.9	2.8%
W08	LED bar return	12	black	7.2	17.1	
W09	K4 to searchlight	16	red	3.7	17.3	3.7%
W10	Searchlight return	16	black	3.7	17.4	
W11	K2 to stern light	16	dark gray	0.5	17.7	0.5%
W12	Stern light return	16	black	0.5	17.9	
W13	K2 to port side light	16	dark gray	0.25	5.3	0.1%
W14	Port side light return	16	black	0.25	5.2	
W15	F5 to dash switch bus	16	orange	6	5.6	2.0%
W16	K1 to dash accessory bus	16	purple	3.5	5.8	1.2%
W17	Dash accessory return	16	black	3.5	5.7	
W18	F7 to bilge pump	16	brown	2	7.5	0.9%
W19	Bilge pump return	16	black	2	7.7	
W20	K5 to fuel pump	12	brown	6	9.9	1.4%
W21	Fuel pump return	12	black	6	9.8	
W22	Switch bus to key	16	orange	6	0.8	
W23	Switch bus to lights switch	16	orange	1	1.0	
W24	Switch bus to searchlight switch	16	orange	1	1.2	
W25	Accessory bus to USB	16	orange	1.5	1.5	
W26	USB return	16	black	1.5	1.5	
W27	Accessory bus to gas gauge	16	purple	0.3	1.3	
W28	Gas gauge return	16	black	0.3	1.3	
W29	Accessory bus to tach	16	purple	0.3	1.0	
W30	Tach return	16	black	0.3	1.0	
W31	Accessory bus to temperature gauge	16	purple	0.3	0.8	
W32	Temperature gauge return	16	black	0.3	0.8	
W33	Accessory bus to exhaust gauge	16	purple	0.3	0.7	
W34	Exhaust gauge return	16	black	0.3	0.7	
W35	Key ON to K1 coil and engine	16	purple	0.5	11.7	
W36	Key START to solenoid	16	yellow/red	5	11.5	
W37	Lights switch RUN to K2 coil	16	dark gray	0.2	6.2	
W38	Lights switch LED to K3 coil	16	gray/blue	0.2	6.2	
W39	Searchlight switch to K4 coil	16	gray/red	0.2	6.4	
W40	Engine pump signal to K5 coil	16	brown/white	0.2	7.0	
W41	Tach signal	16	dark gray	0.05	12.1	
W42	Temperature sender signal	16	tan	0.05	12.8	
W43	Temperature sender return	16	black	0.05	12.8	
W44	Gas sender signal	16	pink	0.05	13.6	
W45	Gas sender return	16	black	0.05	13.6	
W46	Exhaust probe, type K extension	18	K yellow	0.0	14.6	
W47	DESS post to ECM, wire 1	18	white	0.05	12.6	
W48	DESS post to ECM, wire 2	18	white/blue	0.05	12.5	
W49	DESS post to ECM, wire 3	18	white/green	0.05	12.5	
W50	Searchlight remote cable	18	gray	0.1	22.6	
W51	Hub fuse to relay K1 accessory	16	red	3.5	1.4	
W52	Hub fuse to relay K2 navigation	16	dark gray	1	1.4	
W53	Hub fuse to relay K3 LED bar	12	red	7.2	1.4	
W54	Hub fuse to relay K4 searchlight	16	red	3.7	1.4	
W55	Hub fuse to relay K5 fuel pump	12	brown	6	1.4	
W56	Relay coil returns (5, daisy chain)	16	black	1	1.4	
W57	Battery positive to battery switch	2	red	150	1.1	
W58	K2 to starboard side light	16	dark gray	0.25	5.7	
W59	Starboard side light return	16	black	0.25	5.6	
W60	Low oil sensor to dash LED	16	white/blue	0.02	12.6	
W61	Low oil sensor return	16	black	0.02	12.4	
W62	Accessory bus to low oil LED	16	purple	0.02	0.7	

Fuses

MF1 60 A Main feed, on the battery switch output stud (W03)
F1 20 A 12 V outlet (W05)
F2 10 A LED bar (K3) (W07)
F3 10 A Searchlight (K4) (W09)
F4 10 A Navigation lights (K2) (W11, W13, W58)
F5 10 A Dash switch bus: key, lights, searchlight (W15)
F6 10 A Accessory bus: gauges, USB (K1) (W16)
F7 5 A Bilge pump, inline at the battery, ahead of the switch (W18)
F8 15 A Fuel pump (K5) (W20)

Connectors

Battery, solenoid, ground stud, fuse block: sealed ring terminals, adhesive heat shrink.
Relays and dash switches: sealed 1/4 in female spades.
Devices outside the dry box: Deutsch DT, 2-pin for a device pair. Engine J1: Deutsch DT 12-pin on the stand.
Gauges: ring terminals on the gauge studs. Exhaust probe: type K mini plug.
TB1 and TB2: screw-terminal bars, one ring terminal on each screw (E-11 limit: 4 on a stud).

To confirm from the 800R service manual

Engine J1 pinout: ignition on, tach, fuel pump output, DESS post.
Whether the ECM runs the pump on and off, which K5 assumes, or varies it.
Fuel pump current. 6 A is an estimate.
Charging output and where the regulator lands.
Solenoid coil current. 5 A is an estimate. Whether the sled starter has its own solenoid.
Whether the kept DESS post works as the lanyard kill.
Oil level sensor: pinout, and that it closes on low oil. Oil pump wiring.